Universidad de las Américas Puebla

Escuela de ingeniería

Departamento de computación, electrónica y mecatrónica

Lab report #3

Course: Digital design LRT2022-7

Equipo: 2

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1 Abstract

This report will present the results obtained from the third laboratory practice of the Digital Design course. The main objective of this practice was to continue to familiarize ourselves with the functioning of logic gates and boolean functions. Showcasing both a program in VHDL and simulation function based on logic gates.

2 Introduction

In this lab, we will study some basic compounded gates after learning about the most fundamental logic gates. To carry out more intricate operations, these gates are made by combining several simple logic gates. We can start creating increasingly complex circuits that can perform a greater variety of tasks by comprehending how these compounded gates operate. This will enable us to increase our understanding of and proficiency in the field of electronics while also delving deeper into the realm of digital logic.

- NAND, output is false only if all inputs are true.
- NOR, output is true only if any input is false.
- XOR, outputs true if inputs are different.
- XNOR, outputs true if inputs are the same.

[1] It's critical to comprehend not only how these compounded gates operate but also the various ways in which they can be connected within a circuit. Additionally, we can investigate the plethora of possibilities in the field of digital logic by experimenting with various compounded gate configurations.

3 Objectives

The objectives of this lab are:

- Obtain Boolean functions from schematics.
- Obtain schematics from Boolean functions.
- Program and simulate Boolean functions using VHDL.

For the purpose of this lab, it is important to know about the components used, in the first place, "logic gates are defined as simple digital circuits that take one or more binary inputs and produce a binary output"[2]. On the other hand, boolean funtions consists of a number of Boolean variables joined by the Boolean connectives, like AND and OR. [3] Similary to the previous lab, the software utilized in the experiment is VHDL (Hardware Description Language), which is a language that describes the behavior of electronic circuits, most commonly digital circuits. [4] For this experiment, it is fundamental to explain the procedure to convert a schematic into a boolean function, in this case, one must know how to write a truth table, and construct the canonic form of the function.

To understand this, we must define the elements of a electronic schematic. For most circuits, a photograph or realistic drawing showing actual components and their interconnections would be far too complex to be of value in replicating the circuit. A schematic symbol is a simplified representation of a real-world component. A schematic diagram shows such representations of real-world components and a simplified "map" of how they are connected together.[5]
Once a schematic is created or obtained, the truth table must be generated, since the truth table is defined as a canonical representation of a Boolean function that exhaustively enumerates every truth assignment and its corresponding function value. [6]
And it is through this truth table that a canonic function can be obtained, which refers to a Boolean function that is expressed either as a sum of minterms or as a product of maxterms. It represents the standard form of a Boolean function that can be manipulated to simplify its expression. [7]

4 Methodology

A methodology was established, divided into three main phases: physical implementation in the laboratory, simulation in Multisim, and programming in EDA Playground.

4.1 Physical Laboratory Phase

In the physical laboratory, the following procedure was carried out:

- The power supply was configured with the following parameters:
  - Activated source: Source 3.
  - Output voltage: 5 V.
  - Output current: 300 mA.
- Using the datasheets, the power terminals, inputs, and outputs of each logic gate were identified.
- The basic circuit of each logic gate was built.
- The truth table presented in the datasheet was verified and compared with the one obtained from the real circuit.
- The truth table and the canonical form of the Boolean function were obtained:

$$F(A, B, C, D) = (\bar{B} \cdot D) + (\bar{A} \cdot B \cdot \bar{C}) + (A \cdot \bar{B} \cdot C) + (A \cdot B \cdot \bar{C})$$

- The circuit corresponding to the Boolean function was built.
- The truth table of the Boolean function was verified against the one obtained experimentally from the real circuit.

4.2 Multisim Simulation Phase

In the Multisim environment, the following tasks were performed:

- The circuits corresponding to the basic logic gates (AND, OR, XOR, NOT, NAND, NOR, and XNOR) were constructed, simulated, and their correct operation was verified.
- The Boolean function defined in point 5 of the laboratory phase was constructed and simulated, confirming its proper functioning.

4.3 EDA Playground Programming Phase

Finally, in the EDA Playground platform, the following steps were carried out:

- A single file was created to program and verify the correct operation of the basic logic gates (AND, OR, XOR, NOT, NAND, NOR, and XNOR).
- The Boolean function F(ABCD) was programmed and simulated.

Finally, an analysis was carried out between the results obtained in the simulation and those obtained experimentally in the laboratory.

5 Results

5.1 Physical lab experimentation

In this lab, the physical circuits were not built, but based on a schematic, the team developed a truth table and a function:+

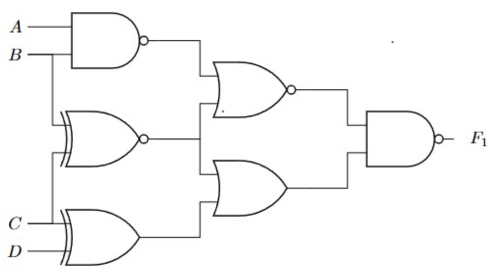

Figure 1: Schematic

Figure 1: Schematic 1, from which the truth table and function are based

After analysing this circuit, the following function was formulated:

$$Y = \overline{(\overline{(\overline{A \cdot B}) + \overline{(B \oplus C)}}) \cdot (\overline{(B \oplus C)} + (C \oplus D)))}$$

And, based on both the function and the schematic, the following truth table was generated:

Input				Output
A	**B**	**C**	**D**	**Y**
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

Table 1: Truth Table for the first schematic provided

And, after developing the code in EDA Playgrounds, the following Wave was generating, showcasing the values of the function:

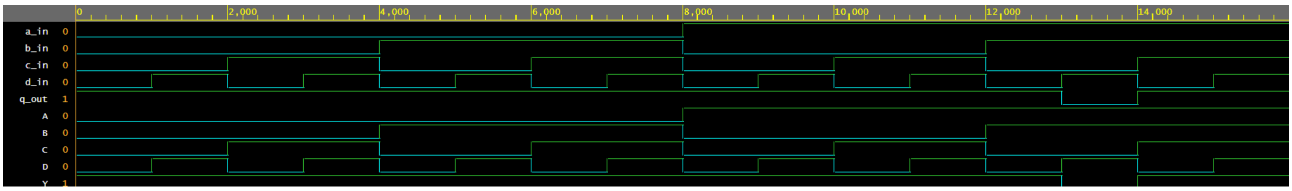

Figure 2: Wave made from the function of the Schematic 1.

For the second experiment, the team was given a function, with the mission of obtaining the schematic and truth table:

$$F(A, B, C, D) = (\bar{B} \cdot D) \cdot (A \oplus B) + (\overline{C \cdot A})$$

And after analysing the function, the following schematic was generated

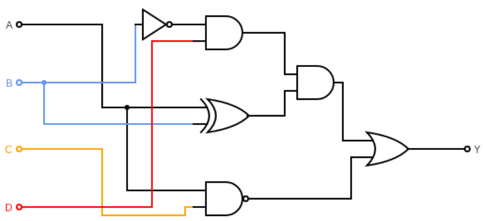

Figure 3: Schematic 2, based on the function given.

And, based on both the function and the schematic, the following truth table was generated:

Input				Output
A	**B**	**C**	**D**	**Y**
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

Table 2: Truth Table for the first schematic provided

And, after developing the code in EDA Playgrounds, the following Wave was generating, showcasing the values of the function:

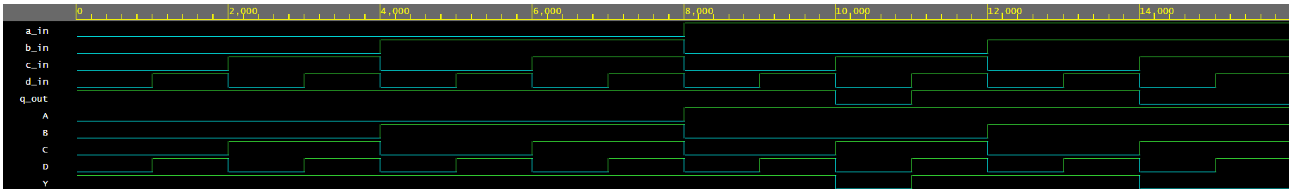

Figure 4: Wave made from the function of the Schematic 2.

5.2 Simulation

Based on the schematics of the previous experiments, the following simulation was built on MultiSim:

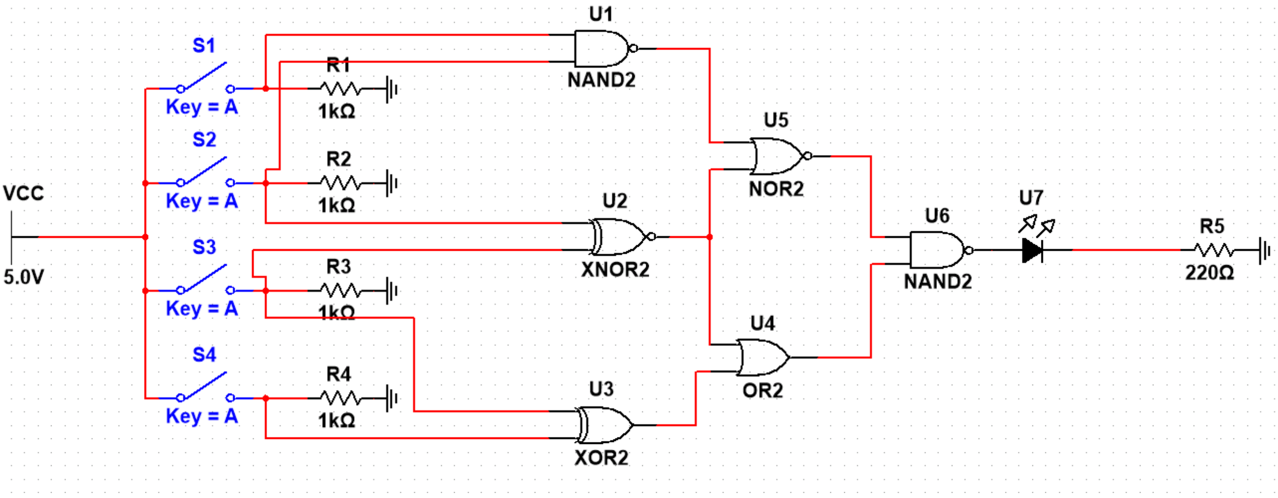

Figure 5: MultiSim schematics made for the first schematic.

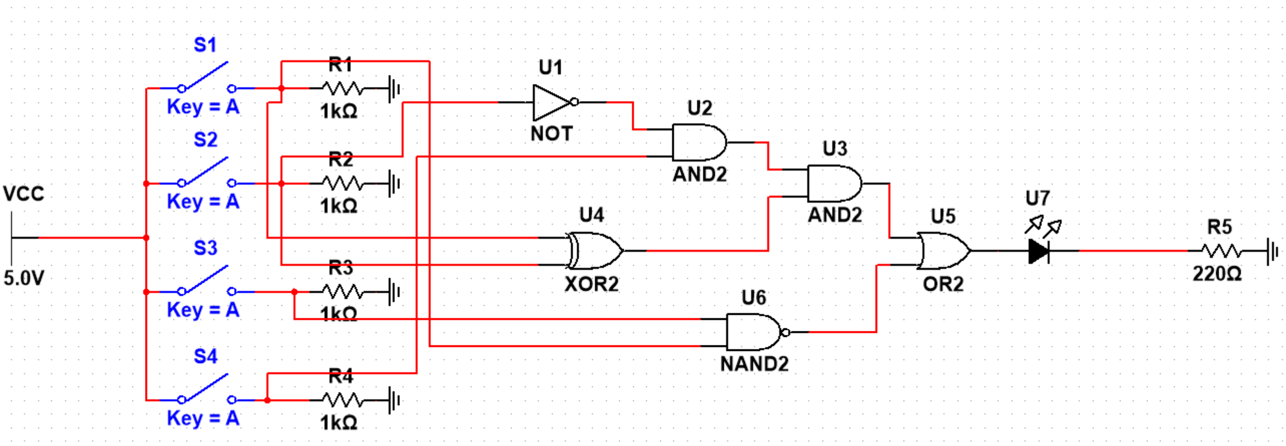

Figure 6: MultiSim schematics made for the second schematic.

5.3 Codes

Finally, in order to generate the previous Waves and to prove the truth tables, the team created the following codes:

Código 1: Code 1: For exercise 1

```
library IEEE;
use IEEE.std_logic_1164.all;

-- Entity declaration

entity ent_lab_3 is

    port(A : in std_logic;        -- OR gate input
         B : in std_logic;
```

```
        C : in std_logic;
        D : in std_logic;
        Y : out std_logic);     -- OR gate output

end ent_lab_3;

-- Architecture definition

architecture arq_lab_3 of ent_lab_3 is

 begin

    Y <= (((A nand B)nor(B xnor C))nand((B xnor C)or(C xor D)));

end arq_lab_3;
```

Código 2: Code 2: For exercise 2

```
library IEEE;
use IEEE.std_logic_1164.all;

-- Entity declaration

entity ent_lab_3_2 is

    port(A : in std_logic;
         B : in std_logic;
         C : in std_logic;
         D : in std_logic;
         Y : out std_logic);

end ent_lab_3_2;

-- Architecture definition

architecture arq_lab_3_2 of ent_lab_3_2 is

 begin

    Y <= (((not(B)and D)and(A xor b))or(C nand A));

end arq_lab_3_2;
```

And to test this codes, the team created the following:

Código 3: Code 1 and 2 Testbench

```
library IEEE;
```

```
use IEEE.std_logic_1164.all;

entity testbench is
-- empty
end testbench;

architecture tb of testbench is

-- DUT component
component ent_lab_3_2 is
port(
  A: in std_logic;
  B : in std_logic;
  C : in std_logic;
  D : in std_logic;
  Y: out std_logic);
end component;

signal a_in, b_in, c_in, d_in, q_out: std_logic;

begin

  -- Connect UUT
  UUT: ent_lab_3_2 port map(a_in, b_in, c_in, d_in, q_out);

  process
  begin

    a_in <= '0';
    b_in <= '0';
    c_in <= '0';
    d_in <= '0';
    wait for 1 ns;

    a_in <= '0';
    b_in <= '0';
    c_in <= '0';
    d_in <= '1';
    wait for 1 ns;

    a_in <= '0';
    b_in <= '0';
    c_in <= '1';
    d_in <= '0';
    wait for 1 ns;
```

```
a_in <= '0';
b_in <= '0';
c_in <= '1';
d_in <= '1';
wait for 1 ns;

a_in <= '0';
b_in <= '1';
c_in <= '0';
d_in <= '0';
wait for 1 ns;

a_in <= '0';
b_in <= '1';
c_in <= '0';
d_in <= '1';
wait for 1 ns;

a_in <= '0';
b_in <= '1';
c_in <= '1';
d_in <= '0';
wait for 1 ns;

a_in <= '0';
b_in <= '1';
c_in <= '1';
d_in <= '1';
wait for 1 ns;

a_in <= '1';
b_in <= '0';
c_in <= '0';
d_in <= '0';
wait for 1 ns;

a_in <= '1';
b_in <= '0';
c_in <= '0';
d_in <= '1';
wait for 1 ns;

a_in <= '1';
b_in <= '0';
```

```
    c_in <= '1';
    d_in <= '0';
    wait for 1 ns;

    a_in <= '1';
    b_in <= '0';
    c_in <= '1';
    d_in <= '1';
    wait for 1 ns;

    a_in <= '1';
    b_in <= '1';
    c_in <= '0';
    d_in <= '0';
    wait for 1 ns;

    a_in <= '1';
    b_in <= '1';
    c_in <= '0';
    d_in <= '1';
    wait for 1 ns;

    a_in <= '1';
    b_in <= '1';
    c_in <= '1';
    d_in <= '0';
    wait for 1 ns;

    a_in <= '1';
    b_in <= '1';
    c_in <= '1';
    d_in <= '1';
    wait for 1 ns;

  end process;
end tb;
```

6 Analysis

Based on the data obtained during the laboratory practice, the observed behavior of the logic gates corresponds correctly to the expected truth tables. The discrepancies that initially appeared in the practical implementation were due to wiring errors and circuit organization, which were later corrected. After completing the laboratory, the circuits were reproduced in Multisim to ensure proper organization and eliminate any possible mistakes. Each logic gate was tested individually, confirming its correct functionality. Finally, the circuits were implemented in EDA Playground using VHDL, where their behavior was simulated and verified. The results in this environment also matched the theoretical expectations, validating the practical and simulated work. In conclusion, both the laboratory implementation and the digital simulations in Multisim and VHDL confirm the correct functionality of the logic gates and the designed circuits.

7 Conclusions

The development of this practice made it possible to implement and verify the operation of three basic logic gates, both in the laboratory and in the simulation environments. The use of the multimeter was essential to confirm the correct functioning of the digital circuits, strengthening the understanding of measurement and verification processes in electronics. Additionally, the Boolean function was analyzed by obtaining its truth table and canonical form, which was then implemented and validated experimentally. This analysis was complemented with simulations in Multisim and programming in VHDL through EDA Playground, confirming the correct logical behavior of the circuits and the function in different environments. In this way, the proposed objectives were satisfactorily achieved, as the laboratory practice not only allowed the experimental validation of theoretical principles but also demonstrated the usefulness of digital simulation tools such as Multisim and VHDL for predicting and verifying circuit behavior.

References

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